BT613 T - Agriculture Biotechnology

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Assignment No. 02

Spring 2024
Due Date: July 03, 2024

Assignment Topic:

Imagine a plant is facing any type of biotic stress. How the plant will manage to survive under stress condition. Take any example of your choice (any biotic stress) and explain the defence mechanism that plant will adopt against that biotic stress.
Note: You can choose any biotic stress factor e.g. Bacteria, Virus, Fungus, Nematodes, Insect-pests etc. Your answer should be according to chosen stress factor.

Ans

Plant Defense Mechanisms Against Fungal Pathogens

I. Introduction

Plants are constantly exposed to various biotic stresses, including fungal pathogens. Fungi can cause significant damage to plants, leading to reduced crop yields and economic losses. However, plants have evolved sophisticated defense mechanisms to combat these threats. This document outlines the multi-layered defense strategies plants employ against fungal pathogens.

II. Preformed Defenses (Constitutive)

Before a fungal pathogen even attempts to infect, plants have several preexisting barriers:

1. Physical Barriers:

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- **Waxy cuticle**: Provides a hydrophobic layer that prevents water accumulation on the leaf surface, inhibiting fungal spore germination.
- **Cell walls**: Composed of cellulose, hemicellulose, and pectin, creating a tough barrier against fungal penetration.
- **Bark**: In woody plants, offers protection against many pathogens.

2. Chemical Barriers:

- Phytoanticipins: Preformed antimicrobial compounds like saponins, cyanogenic glucosides, and glucosinolates.
- Inhibitory proteins: Some plants constitutively produce antifungal proteins and enzymes.

III. Induced Defenses (Active Responses)

When a fungal pathogen overcomes preformed defenses, plants activate various induced defense mechanisms:

1. Pathogen Recognition:

- **Pattern Recognition Receptors (PRRs):** Detect Pathogen-Associated Molecular Patterns (PAMPs) like chitin from fungal cell walls.
- **Resistance (R) proteins:** Recognize specific fungal effectors, triggering effector-triggered immunity (ETI).

2. Signaling Cascades:

- **Mitogen-Activated Protein Kinase (MAPK) cascades**: Rapidly transmit signals from receptors to the nucleus.
- **Hormone signaling**: Salicylic acid (SA), jasmonic acid (JA), and ethylene play crucial roles in defense response regulation.

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3. Hypersensitive Response (HR):

- Rapid, localized cell death around the infection site to contain the pathogen.
- Often associated with the production of reactive oxygen species (ROS).

4. Structural Defenses:

- **Callose deposition**: Forms physical barriers at infection sites.
- **Lignification**: Strengthens cell walls to resist fungal penetration.

5. Biochemical Defenses:

- **Phytoalexins**: Antimicrobial compounds synthesized in response to infection (e.g., camalexin in Arabidopsis).
- **Pathogenesis**-Related (PR) proteins: Various proteins with antifungal activities, including chitinases and glucanases that can degrade fungal cell walls.

6. Systemic Acquired Resistance (SAR):

- Long-lasting, broad-spectrum resistance throughout the plant.
- Mediated by salicylic acid and the accumulation of PR proteins.

IV. Specific Example: Wheat Defense Against Fusarium graminearum

Fusarium graminearum causes Fusarium Head Blight (FHB) in wheat. Here's how wheat responds:

1. Recognition:

- Wheat PRRs recognize chitin fragments from F. Graminearum cell walls.
- R proteins may recognize specific F. Graminearum effectors.

2. Early Responses:

- Activation of MAPK cascades and calcium signaling.

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- Rapid production of reactive oxygen species (ROS).

3. Hormonal Regulation:

- Increase in salicylic acid and jasmonic acid levels.
- Ethylene production contributes to defense gene activation.

4. Gene Expression Changes:

- Upregulation of genes involved in phenylpropanoid pathway (for lignin synthesis).
- Increased expression of PR genes, including chitinases and β-1,3-glucanases.

5**. Metabolic Adjustments:**

- Production of antimicrobial compounds like benzoxazinoids.
- Synthesis of phenolic compounds and lignin precursors.

6. Structural Modifications:

- Callose deposition at infection sites.
- Cell wall fortification through increased lignification.

7. Systemic Responses:

- Induction of SAR, providing enhanced resistance to subsequent infections.

V. Conclusion

Plants employ a multi-layered, dynamic defense system against fungal pathogens. This system involves both preformed barriers and induced responses, utilizing physical, chemical, and molecular mechanisms. Understanding these defense mechanisms is crucial for developing strategies to enhance crop resistance to fungal diseases, ultimately improving agricultural productivity and food security.

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General instruction:

Dear Students:

Please note this assignment is "Inline" assignment and copy paste is not allowed. You will write your answers within given space.

- The length of assignment should be 700-1000 words.
- Give the title at top of the assignment with proper subheadings.
- Please be specific and to the point and give the answer according to question avoiding irrelevant details.
- This is not a group assignment; it is an individual assignment so be careful and avoid copying others' work.
- Give references where necessary.

- This assignment carries 10 marks.
- Assignment should be submitted on LMS not later than **July 03, 2024.**

DO NOT send the assignment thorough email. Assignments submitted through email will result in no marks.

Outcomes:

This assignment will help you to understand the plant's response against biotic stresses.

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